

FIDE Swiss Rules

C.04.1 Basic rules for Swiss Systems

The following rules are valid for each Swiss system unless explicitly stated otherwise.	
a	The number of rounds to be played is declared beforehand.
b	Two players shall not play against each other more than once.
c	Should the number of players to be paired be odd, one player is unpaired. This player receives a pairing-allocated bye: no opponent, no colour and as many points as are rewarded for a win, unless the rules of the tournament state otherwise.
d	A player who has already received a pairing-allocated bye, or has already scored a (forfeit) win due to an opponent not appearing in time, shall not receive the pairing-allocated bye.
e	In general, players are paired to others with the same score.
f	For each player the difference between the number of black and the number of white games shall not be greater than 2 or less than -2. Each system may have exceptions to this rule in the last round of a tournament.
g	No player shall receive the same colour three times in a row. Each system may have exceptions to this rule in the last round of a tournament.
h	1. In general, a player is given the colour with which he played less games.
	2. If colours are already balanced, then, in general, the player is given the colour that alternates from the last one with which he played.
i	The pairing rules must be such transparent that the person who is in charge for the pairing can explain them.

C.04.2 General handling rules for Swiss Tournaments

A.	Pairing Systems	
	1	The pairing system used for a FIDE rated Swiss tournament should be one of the published FIDE Swiss Systems. Accelerated methods are acceptable if they were announced in advance by the organizer and are published in section C.04.5.
	2	In derogation of the previous rule, unpublished pairing systems or accelerated methods may be permitted, provided that a detailed written description of their rules: 1. be submitted in advance to the Qualification Commission (QC) and temporarily authorized by them; and 2. be explicitly presented to the participants before the start of the tournament.
	3	While reporting a tournament to FIDE, the Arbiter shall declare which official FIDE Swiss system and acceleration method (if any) were used, or provide the temporary authorization(s) given by the QC as per the previous rule.
	4	The Swiss Pairing Systems defined by FIDE and not deprecated (see C.04.4) pair the players in an objective, impartial and reproducible way.

		In any tournament where such systems are used, different arbiters, or different endorsed software programs, must be able to arrive at identical pairings.
	5	It is not allowed to alter the correct pairings in favour of any player. Where it can be shown that modifications of the original pairings were made to help a player achieve a norm or a direct title, a report may be submitted to the QC to initiate disciplinary measures through the Ethics Commission.
B.	Initial Order	
	1	Before the start of the tournament, a measure of the player's strength is assigned to each player. The strength is usually represented by rating lists of the players. If one rating list is available for all participating players, then this rating list should be used. It is advisable to check all ratings supplied by players. If no reliable rating is known for a player, the arbiters should make an estimation of it as accurately as possible.
	2	Before the first round the players are ranked in order of, respectively 1. Strength (rating) 2. FIDE-title (GM-IM- WGM-FM-WIM-CM-WFM-WCM-no title) 3. alphabetically (unless it has been previously stated that this criterion has been replaced by another one)
	3	This ranking is used to determine the pairing numbers; the highest one gets #1 etc. If, for any reason, the data used to determine the rankings were not correct, they can be adjusted at any time. The pairing numbers may be reassigned accordingly to the corrections. No modification of a pairing number is allowed after the fourth round has been paired.
C.	Late Entries	
	1	According to FIDE Tournament Rules, any prospective participant who has not arrived at the venue of a FIDE competition before the time scheduled for the drawing of lots shall be excluded from the tournament unless he shows up at the venue in time before a pairing of another round. An exception may be made in the case of a registered participant who has given written notice in advance that he will be unavoidably late.
	2	Where the Chief Arbiter decides to admit a latecomer, • if the player's notified time of arrival is in time for the start of the first round, the player is given a pairing number and paired in the usual way. • if the player's notified time of arrival is in time only for the start of the second (or third) round ("Late Entry"), then the player is not paired for the rounds which he cannot play. Instead, he receives no points for unplayed rounds (unless the rules of the tournament say otherwise), and is given an appropriate pairing number and paired only when he actually arrives.
	3	If there are late entries, the Pairing Numbers that were given at the start of the tournament are considered provisional. The definitive Pairing Numbers are given only when the List of Participants is closed, and corrections made accordingly in the results charts.
D.	Pairing, colour and publishing rules	
	1	Adjourned games are considered draws for pairing purposes only.

	2	A player who is absent without notifying the arbiter will be considered as withdrawn, unless the absence is explained with acceptable arguments before the next pairing is published.
	3	Players who withdraw from the tournament will no longer be paired.
	4	Players known in advance not to play in a particular round are not paired in that round and score zero (unless the rules of the tournament say otherwise).
	5	Only played games count in situations where the colour sequence is meaningful. So, for instance, a player with a colour history of BWB=W (i.e. no valid game in round-4) will be treated as if his colour history was =BWBW. WB=WB will count as =WBWB, BWW=B=W as ==BWWBW and so on.
	6	Two paired players, who did not play their game, may be paired together in a future round.
	7	The results of a round shall be published at the usual place of communication at announced time due to the schedule of the tournament.
	8	If either • a result was written down incorrectly, or • a game was played with the wrong colours, or • a player's rating has to be corrected (and playing numbers possibly recomputed as in C.04.2.C.3), and a player communicates this to the arbiter within a given deadline after publication of results, the new information shall be used for the standings and the pairings of the next round. The deadline shall be fixed in advance according to the timetable of the tournament. If the error notification is made after the pairing but before the end of the next round, it will affect the next pairing to be done. If the error notification is made after the end of the next round, the correction will be made after the tournament for submission to rating evaluation only.
	9	After a pairing is complete, sort the pairs before publishing them. The sorting criteria are (with descending priority) • the score of the higher ranked player of the involved pair; • the sum of the scores of both players of the involved pair; • the rank according to the Initial Order (C.04.2.B) of the higher ranked player of the involved pair.
	10	Once published, the pairings shall not be changed unless they are found to violate C.04.1.b (<i>Two players shall not play against each other more than once</i>).

C.04.3 FIDE (Dutch) System

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Version approved at the 87th FIDE Congress in Baku 2016.		
Terms and Definitions and Pairing Guidelines For Programmers added at the 88th FIDE Congress in Goynuk 2017. See http://pairings.fide.com/documents/fide-dutch-extras.html		
A	Introductory Remarks and Definitions	
	A.1	Initial ranking list
		See C.04.2.B (General Handling Rules – Initial order)
	A.2	Order
		For pairings purposes only, the players are ranked in order of, respectively
	a	score
	b	pairing numbers assigned to the players accordingly to the initial ranking list and subsequent modifications depending on possible late entries or rating adjustments
	A.3	Scoregroups and pairing brackets
		A scoregroup is normally composed of (all) the players with the same score. The only exception is the special “collapsed” scoregroup defined in A.9. A (pairing) bracket is a group of players to be paired. It is composed of players coming from one same scoregroup (called resident players) and of players who remained unpaired after the pairing of the previous bracket. A (pairing) bracket is homogeneous if all the players have the same score; otherwise it is heterogeneous. A remainder (pairing bracket) is a sub-bracket of a heterogeneous bracket, containing some of its resident players (<i>see B.3 for further details</i>).
	A.4	Floater and floats
	a	A downfloater is a player who remains unpaired in a bracket, and is thus moved to the next bracket. In the destination bracket, such players are called “moved-down players” (MDPs for short).
	b	After two players with different scores have played each other in a round, the higher ranked player receives a downfloat, the lower one an upfloat. A player who, for whatever reason, does not play in a round, also receives a downfloat.
	A.5	Byes
		See C.04.1.c (<i>Should the number of players to be paired be odd, one player is unpaired. This player receives a pairing-allocated bye: no opponent, no colour and as many points as are rewarded for a win, unless the regulations of the tournament state otherwise</i>).
	A.6	Colour differences and colour preferences
		The colour difference of a player is the number of games played with white minus the number of games played with black by this player. The colour preference is the colour that a player should ideally receive for

		the next game. It can be determined for each player who has played at least one game.
	a	An absolute colour preference occurs when a player's colour difference is greater than +1 or less than -1, or when a player had the same colour in the two latest rounds he played. The preference is white when the colour difference is less than -1 or when the last two games were played with black. The preference is black when the colour difference is greater than +1, or when the last two games were played with white.
	b	A strong colour preference occurs when a player's colour difference is +1 (preference for black) or -1 (preference for white).
	c	A mild colour preference occurs when a player's colour difference is zero, the preference being to alternate the colour with respect to the previous game he played.
	d	Players who did not play any games have no colour preference (the preference of their opponents is granted).
	A.7	Topscoreers
		Topscoreers are players who have a score of over 50% of the maximum possible score when pairing the final round of the tournament.
	A.8	Pairing Score Difference (PSD)
		The pairing of a bracket is composed of pairs and downfloaters. Its Pairing Score Difference is a list of score-differences (<i>SD, see below</i>), sorted from the highest to the lowest. For each pair in a pairing, the SD is defined as the absolute value of the difference between the scores of the two players who constitute the pair. For each downfloater, the SD is defined as the difference between the score of the downfloater, and an artificial value that is one point less than the score of the lowest ranked player of the current bracket (even when this yields a negative value).
		<i>Note: The artificial value defined above was chosen in order to be strictly less than the lowest score of the bracket, and generic enough to work with different scoring-point systems and in presence of non-existent, empty or sparsely populated brackets that may follow the current one.</i>
		PSD(s) are compared lexicographically (<i>i.e. their respective SD(s) are compared one by one from first to last – in the first corresponding SD(s) that are different, the smallest one defines the lower PSD</i>).
	A.9	Round-Pairing Outlook
		The pairing of a round (called round-pairing) is complete if all the players (except at most one, who receives the pairing-allocated bye) have been paired and the absolute criteria C1-C3 have been complied with. If it is impossible to complete a round-pairing, the arbiter shall decide what to do. Otherwise, the pairing process starts with the top scoregroup, and continues bracket by bracket until all the scoregroups, in descending order, have been used and the round-pairing is complete. However, if, during this process, the downfloaters (possibly none) produced by the bracket just paired, together with all the remaining players, do not allow the completion of the round-pairing, a different processing route is

		followed. The last paired bracket is called Penultimate Pairing Bracket (PPB). The score of its resident players is called the “collapsing” score. All the players with a score lower than the collapsing score constitute the special “collapsed” scoregroup mentioned in A.3. The pairing process resumes with the re-pairing of the PPB. Its downfloaters, together with the players of the collapsed scoregroup, constitute the Collapsed Last Bracket (CLB), the pairing of which will complete the round-pairing.	
		<i>Note: Independently from the route followed, the assignment of the pairing-allocated bye (see C.2) is part of the pairing of the last bracket.</i>	
		Section B describes the pairing process of a single bracket. Section C describes all the criteria that the pairing of a bracket has to satisfy. Section E describes the colour allocation rules that determine which players will play with white.	
B	Pairing Process for a bracket		
	B.1	Parameters definitions	
		a	M0 is the number of MDP(s) coming from the previous bracket. It may be zero.
		b	MaxPairs is the maximum number of pairs that can be produced in the bracket under consideration <i>(see C.5)</i> .
			<i>Note: MaxPairs is usually equal to the number of players divided by two and rounded downwards. However, if, for instance, M0 is greater than the number of resident players, MaxPairs is at most equal to the number of resident players.</i>
		c	M1 is the maximum number of MDP(s) that can be paired in the bracket <i>(see C.6)</i> .
			<i>Note: M1 is usually equal to the number of MDPs coming from the previous bracket, which may be zero. However, if, for instance, M0 is greater than the number of resident players, M1 is at most equal to the number of resident players.</i> <i>Of course, M1 can never be greater than MaxPairs.</i>
	B.2	Subgroups (original composition)	
		To make the pairing, each bracket will be usually divided into two subgroups, called S1 and S2. S1 initially contains the highest N1 players (sorted according to A.2), where N1 is either M1 <i>(in a heterogeneous bracket)</i> or MaxPairs <i>(otherwise)</i>. S2 initially contains all the remaining resident players. When M1 is less than M0, some MDPs are not included in S1. The excluded MDPs <i>(in number of M0 – M1)</i>, who are neither in S1 nor in S2, are said to be in a <i>Limbo</i>.	
		<i>Note:</i>	<i>the players in the Limbo cannot be paired in the bracket, and are thus bound to double-float.</i>
	B.3	Preparation of the candidate	
		S1 players are tentatively paired with S2 players, the first one from S1 with the first one from S2, the second one from S1 with the second one from S2	

		and so on. In a homogeneous bracket: the pairs formed as explained above and all the players who remain unpaired (bound to be downfloaters) constitute a candidate (pairing). In a heterogeneous bracket: the pairs formed as explained above match M1 MDPs from S1 with M1 resident players from S2. This is called a MDP-Pairing. The remaining resident players (<i>if any</i>) give rise to the remainder (<i>see A.3</i>), which is then paired with the same rules used for a homogeneous bracket.
		<i>Note:</i> M1 may sometimes be zero. In this case, S1 will be empty and the MDP(s) will all be in the Limbo. Hence, the pairing of the heterogeneous bracket will proceed directly to the remainder.
		A candidate (pairing) for a heterogeneous bracket is composed by a MDP-Pairing and a candidate for the ensuing remainder. All players in the Limbo are bound to be downfloaters.
	B.4	Evaluation of the candidate
		If the candidate built as shown in B.3 complies with all the absolute and completion criteria (<i>from C.1 to C.4</i>), and all the quality criteria from C.5 to C.19 are fulfilled, the candidate is called “perfect” and is (immediately) accepted. Otherwise, apply B.5 in order to find a perfect candidate; or, if no such candidate exists, apply B.8.
	B.5	Actions when the candidate is not perfect
		The composition of S1, Limbo and S2 has to be altered in such a way that a different candidate can be produced. The articles B.6 (for homogeneous brackets and remainders) and B.7 (for heterogeneous brackets) define the precise sequence in which the alterations must be applied. After each alteration, a new candidate shall be built (<i>see B.3</i>) and evaluated (<i>see B.4</i>).
	B.6	Alterations in homogeneous brackets or remainders
		Alter the order of the players in S2 with a transposition (<i>see D.1</i>). If no more transpositions of S2 are available for the current S1, alter the original S1 and S2 (<i>see B.2</i>) applying an exchange of resident players between S1 and S2 (<i>see D.2</i>) and reordering the newly formed S1 and S2 according to A.2.
	B.7	Alterations in heterogeneous brackets
		Operate on the remainder with the same rules used for homogeneous brackets (<i>see B.6</i>).
		<i>Note:</i> The original subgroups of the remainder, which will be used throughout all the remainder pairing process, are the ones formed right after the MDP-Pairing. They are called S1R and S2R (to avoid any confusion with the subgroups S1 and S2 of the complete heterogeneous bracket).
		If no more transpositions and exchanges are available for S1R and S2R, alter the order of the players in S2 with a transposition (<i>see D.1</i>), forming a new MDP-Pairing and possibly a new remainder (to be processed as written above). If no more transpositions are available for the current S1, alter, if possible (i.e. if there is a Limbo), the original S1 and Limbo (<i>see B.2</i>), applying an exchange of MDPs

		between S1 and the Limbo (<i>see D.3</i>), reordering the newly formed S1 according to A.2 and restoring S2 to its original composition.
	B.8	Actions when no perfect candidate exists
		Choose the best available candidate. In order to do so, consider that a candidate is better than another if it better satisfies a quality criterion (C5-C19) of higher priority; or, all quality criteria being equally satisfied, it is generated earlier than the other one in the sequence of the candidates (<i>see B.6 or B.7</i>).
C	Pairing Criteria	
	Absolute Criteria	
	No pairing shall violate the following absolute criteria:	
	C.1	see C.04.1.b (<i>Two players shall not play against each other more than once</i>)
	C.2	see C.04.1.d (<i>A player who has already received a pairing-allocated bye, or has already scored a (forfeit) win due to an opponent not appearing in time, shall not receive the pairing-allocated bye</i>).
	C.3	non-topscorers (<i>see A.7</i>) with the same absolute colour preference (<i>see A6.a</i>) shall not meet (<i>see C.04.1.f and C.04.1.g</i>).
	Completion Criterion	
	C.4	if the current bracket is the PPB (<i>see A.9</i>): choose the set of downfloaters in order to complete the round-pairing.
	Quality Criteria	
	To obtain the best possible pairing for a bracket, comply as much as possible with the following criteria, given in descending priority:	
	C.5	maximize the number of pairs (<i>equivalent to: minimize the number of downfloaters</i>).
	C.6	minimize the PSD (<i>This basically means: maximize the number of paired MDP(s); and, as far as possible, pair the ones with the highest scores</i>).
	C.7	if the current bracket is neither the PPB nor the CLB (<i>see A.9</i>): choose the set of downfloaters in order first to maximize the number of pairs and then to minimize the PSD (<i>see C.5 and C.6</i>) in the following bracket (<i>just in the following bracket</i>).
	C.8	minimize the number of topscorers or topscorers' opponents who get a colour difference higher than +2 or lower than -2.
	C.9	minimize the number of topscorers or topscorers' opponents who get the same colour three times in a row.
	C.10	minimize the number of players who do not get their colour preference.
	C.11	minimize the number of players who do not get their strong colour preference.
	C.12	minimize the number of players who receive the same downfloat as the previous round.
	C.13	minimize the number of players who receive the same upfloat as the previous round.
	C.14	minimize the number of players who receive the same downfloat as two rounds before.

	C.15	minimize the number of players who receive the same upfloat as two rounds before.
	C.16	minimize the score differences of players who receive the same downfloat as the previous round.
	C.17	minimize the score differences of players who receive the same upfloat as the previous round.
	C.18	minimize the score differences of players who receive the same downfloat as two rounds before.
	C.19	minimize the score differences of players who receive the same upfloat as two rounds before.
D	Rules for the sequential generation of the pairings	
	Before any transposition or exchange take place, all players in the bracket shall be tagged with consecutive in-bracket sequence-numbers (BSN for short) representing their respective ranking order (according to A.2) in the bracket (<i>i.e.</i> 1, 2, 3, 4, ...).	
	D.1	Transpositions in S2
		A transposition is a change in the order of the BSNs (<i>all representing resident players</i>) in S2. All the possible transpositions are sorted depending on the lexicographic value of their first N1 BSN(s), where N1 is the number of BSN(s) in S1 (<i>the remaining BSN(s) of S2 are ignored in this context, because they represent players bound to constitute the remainder in case of a heterogeneous bracket; or bound to downfloat in case of a homogeneous bracket – e.g. in a 11-player homogeneous bracket, it is 6-7-8-9-10, 6-7-8-9-11, 6-7-8-10-11, ..., 6-11-10-9-8, 7-6-8-9-10, ..., 11-10-9-8-7 (720 transpositions); if the bracket is heterogeneous with two MDPs, it is: 3-4, 3-5, 3-6, ..., 3-11, 4-3, 4-5, ..., 11-10 (72 transpositions)</i>).
	D.2	Exchanges in homogeneous brackets or remainders (original S1 ↔ original S2)
		An exchange in a homogeneous brackets (also called a resident-exchange) is a swap of two equally sized groups of BSN(s) (<i>all representing resident players</i>) between the original S1 and the original S2. In order to sort all the possible resident-exchanges, apply the following comparison rules between two resident-exchanges in the specified order (<i>i.e. if a rule does not discriminate between two exchanges, move to the next one</i>). The priority goes to the exchange having:
	a	the smallest number of exchanged BSN(s) (<i>e.g. exchanging just one BSN is better than exchanging two of them</i>).
	b	the smallest difference between the sum of the BSN(s) moved from the original S2 to S1 and the sum of the BSN(s) moved from the original S1 to S2 (<i>e.g. in a bracket containing eleven players, exchanging 6 with 4 is better than exchanging 8 with 5; similarly exchanging 8+6 with 4+3 is better than exchanging 9+8 with 5+4; and so on</i>).
	c	the highest different BSN among those moved from the original S1 to S2 (<i>e.g. moving 5 from S1 to S2 is better than moving 4; similarly, 5-2 is better than 4-3; 5-4-1 is better than 5-3-2; and so on</i>).
	d	the lowest different BSN among those moved from the original S2 to S1 (<i>e.g. moving 6 from S2 to S1 is better than moving 7; similarly, 6-9 is better than 7-8; 6-7-10 is better than 6-8-9; and so on</i>).

	D.3	Exchanges in heterogeneous brackets (original S1 ↔ original Limbo)	
		An exchange in a heterogeneous bracket (also called a MDP-exchange) is a swap of two equally sized groups of BSN(s) (<i>all representing MDP(s)</i>) between the original S1 and the original Limbo. In order to sort all the possible MDP-exchanges, apply the following comparison rules between two MDP-exchanges in the specified order (<i>i.e. if a rule does not discriminate between two exchanges, move to the next one</i>) to the players that are in the new S1 after the exchange. The priority goes to the exchange that yields a S1 having:	
		a	the highest different score among the players represented by their BSN (<i>this comes automatically in complying with the C.6 criterion, which says to minimize the PSD of a bracket</i>).
		b	the lowest lexicographic value of the BSN(s) (sorted in ascending order).
	Any time a sorting has been established, any application of the corresponding D.1, D.2 or D.3 rule, will pick the next element in the sorting order.		
E	Colour Allocation rules		
	<i>Initial-colour</i>		
	It is the colour determined by drawing of lots before the pairing of the first round.		
	For each pair apply (with descending priority):		
	E.1	Grant both colour preferences.	
	E.2	Grant the stronger colour preference. If both are absolute (topscorers, see A.7) grant the wider colour difference (see A.6).	
	E.3	Taking into account C.04.2.D.5, alternate the colours to the most recent time in which one player had white and the other black.	
	E.4	Grant the colour preference of the higher ranked player.	
	E.5	If the higher ranked player has an odd pairing number, give him the initial-colour; otherwise give him the opposite colour.	
		<i>Note: Always consider sections C.04.2.B/C (Initial Order/Late Entries) for the proper management of the pairing numbers.</i>	

C.04.4 Other FIDE-approved Pairing Systems

C.04.4.1 Dubov System

<i>Approved by the 2018 General Assembly.</i>		
Preface:		
<i>The Dubov Swiss Pairing System is designed to maximise the fair treatment of the players. This means that a player having more points than another player during a tournament should have a higher performance rating as well.</i> <i>If the average rating of all players is nearly equal, like in a round robin tournament, the goal is reached. As a Swiss System is a statistical system, this goal can only be reached approximately. The approach is the attempt to equalise the average rating of the opponents (ARO, see A.6) of all players of a scoregroup. Therefore, the pairing of a round will now pair players who have a low ARO against opponents having high ratings.</i>		
A	Introductory Remarks and Definitions	
	A.1	Rating Each player must have a rating. If a player does not have a rating, a provisional one must be assigned to the player by the arbiter.
	A.2	Initial ranking list See C.04.2.B (General Handling Rules – Initial order) Each time a player's rating is introduced or modified before the pairing of the fourth round, the arbiter must re-sort the initial ranking list according to the aforementioned section.
	A.3	Scoregroups and pairing brackets A scoregroup is composed of all the players with the same score. A (pairing) bracket is a group of players to be paired. It is composed of players coming from the same scoregroup (called resident players) and (possibly) of players coming from lower scoregroups (called upfloaters). <i>Note: Unlike other systems, there are no downfloaters in the Dubov System.</i>
	A.4	Byes See C.04.1.c (Should the number of players to be paired be odd, one player is unpaired. This player receives a pairing-allocated bye: no opponent, no colour and as many points as are rewarded for a win, unless the regulations of the tournament state otherwise).
	A.5	Colour differences and colour preferences The colour difference of a player is the number of games played with white minus the number of games played with black by this player. The colour preference (<i>also called: due colour</i>) is the colour that a player should ideally receive for the next game. 1. An absolute colour preference occurs when a player's colour difference is greater than +1 or less than -1, or when a player had the same colour in the two latest rounds he played. The preference is white when the colour difference is less than -1 or when the last two games were played with black. The preference is black when the colour difference is greater than +1, or when the last two games were played with white.

		2. A strong colour preference occurs when a player's colour difference is +1 (preference for black) or -1 (preference for white). 3. A mild colour preference occurs when a player's colour difference is zero, the preference being to alternate the colour with respect to the previous game he played. 4. Players who did not play any games are considered to have a mild colour preference for black.
	A.6	Average Rating of Opponents (ARO) ARO is defined for each player who has played at least one game. It is given by the sum of the ratings of the opponents the player met over-the-board (<i>i.e. only played games are used to compute ARO</i>), divided by the number of such opponents, and rounded to the nearest integer number (the higher, if the division ends for 0.5). ARO is computed for each player after each round as a basis for the pairings of the next round. If a player has yet to play a game, his ARO is zero.
	A.7	Maximum upfloater A player is said to be a maximum upfloater when he has already been upfloated a maximum number of times (MaxT). MaxT is a parameter whose value depends on the number of rounds in the tournament (Rnds), and is computed with the following formula: MaxT = 2 + [Rnds/5] where [Rnds/5] means Rnds divided by 5 and rounded downwards.
	A.8	Round-Pairing OutlookThe pairing of a round (called round-pairing) is complete if all the players (except at most one, who receives the pairing-allocated bye) have been paired and the absolute criteria C1-C3 have been complied with. The pairing process starts with the assignment of the pairing-allocated-bye (<i>see B.0</i>) and continues with the pairing of all the scoregroups (<i>see B.1</i>), in descending order of score, until the round-pairing is complete. If it is impossible to complete a round-pairing, the arbiter shall decide what to do. Section B describes the pairing procedures. Section C defines all the criteria that the pairing of a bracket has to satisfy (in order of priority). Section E defines the colour allocation rules that determine which players will play with White.
B	Pairing Procedures	
	Pairing-Allocated-Bye assignment	
	B.0	The pairing-allocated-bye is assigned to the player who: 1. has neither received a pairing-allocated-bye, nor scored a (forfeit) win in the previous rounds (<i>see C.2</i>) 2. allows a complete pairing of all the remaining players (<i>see C.4</i>) 3. has the lowest score 4. has played the highest number of games 5. occupies the lowest position in the initial ranking list (<i>see A.2</i>)
	Pairing Process for a bracket	

	B.1 Determine the minimum number of upfloaters needed to obtain a legal pairing of all the (remaining) resident players of the scoregroup. <i>Note: A pairing is legal when the criteria C.1, C.3 and C.4 are complied with.</i>				
	B.2 Choose the first set of upfloaters (first in the order given by rule D.1) that, together with the (remaining) resident players of this scoregroup, produces a pairing that complies at best with all the pairing criteria (C.1 to C.10). <i>Note: In order to choose the best set of upfloaters, consider that the ensuing bracket (residents + upfloaters) is paired better than another one if it better satisfies a quality criterion (C.5-C.10) of higher priority.</i>				
	B.3 The players of the bracket are divided in two subgroups: <table border="1" data-bbox="350 646 1369 968"> <tr> <td data-bbox="350 646 456 863">G1</td><td data-bbox="456 646 1369 863">This subgroup initially contains the players who have a colour preference for White, unless all the players in the bracket have yet to play a game (<i>like, for instance, in the first round</i>). In the latter case, this subgroup contains the first half of the players of the bracket (according to the initial ranking list).</td></tr> <tr> <td data-bbox="350 863 456 968">G2</td><td data-bbox="456 863 1369 968">This subgroup initially contains the remaining players of the bracket.</td></tr> </table>	G1	This subgroup initially contains the players who have a colour preference for White, unless all the players in the bracket have yet to play a game (<i>like, for instance, in the first round</i>). In the latter case, this subgroup contains the first half of the players of the bracket (according to the initial ranking list).	G2	This subgroup initially contains the remaining players of the bracket.
G1	This subgroup initially contains the players who have a colour preference for White, unless all the players in the bracket have yet to play a game (<i>like, for instance, in the first round</i>). In the latter case, this subgroup contains the first half of the players of the bracket (according to the initial ranking list).				
G2	This subgroup initially contains the remaining players of the bracket.				
	B.4 If players from the smaller subgroup (or from G1, if their sizes are equal) must unavoidably be paired together, a number of players equal to the number of such pairs must be shifted from that subgroup into the other one. Find the <i>*best*</i> set of such players and proceed with the shift. Now, if the number of players in (the possibly new) G1 is different from the number of players in (the possibly new) G2, in order to equalize the size of the two subgroups, extract the <i>*best*</i> set of players from the larger subgroup, and shift those players into the smaller subgroup. <i>Note: *Best*, in both instances, means the first set of players (first in the order given by rule D.2) that can yield a legal pairing that complies at best with C.7.</i>				
	B.5 Sort the players in (the possibly new) G1 in order of ascending ARO or, when AROs are equal, according to the initial ranking list – highest initial ranking first and so on. S1 is the subgroup resulting from such sorting. <i>Note: The sorting of G2 players is described in D.3.</i>				
	B.6 Choose T2, which is the first such transposition of G2 players (transpositions are sorted by rule D.3) that can yield a legal pairing, according to the following generation rule: the first player of S1 is paired with the first player of T2, the second player of S1 with the second player of T2, and so on.				
C	Pairing Criteria				

Absolute Criteria No pairing shall violate the following absolute criteria:	
C.1	see C.04.1.b (<i>Two players shall not play against each other more than once</i>)
C.2	see C.04.1.d (<i>A player who has already received a pairing-allocated bye, or has already scored a (forfeit) win due to an opponent not appearing in time, shall not receive the pairing-allocated bye</i>).
C.3	two players with the same absolute colour preference (see A.5.a) shall not meet (see C.04.1.f and C.04.1.g).
Completion Criterion	
C.4	choose the set of upfloaters in order to complete the round-pairing.
Quality Criteria To obtain the best possible pairing for a bracket, comply as much as possible with the following criteria, given in descending priority:	
C.5	minimize the number of upfloaters.
C.6	minimize the score differences in the pairs involving upfloaters, i.e. maximize the lowest score among the upfloaters (and then the second lowest, and so on).
C.7	minimize the number of players who do not get their colour preference.
C.8	unless it is the last round, minimize the number of upfloaters who are maximum upfloaters (see A.7).
C.9	unless it is the last round, minimize the number of times a maximum upfloater is upfloated.
C.10	unless it is the last round, minimize the number of upfloaters who upfloated in the previous round.
D	Sorting criteria
D.0	Generalities <i>In the articles of this section, the schema below is followed:</i> 1. A pool of P players is selected. 2. Each player in the pool is assigned a sequence number (from #1 to #P) according to a primary sorting criterion. 3. In order to select a set of K such players, the sets will usually be sorted depending on the sequence numbers of their members, put in lexicographic order (exception is D.1.b). For instance, with K=2, the set {#1,#2} will precede {#1,#3}, the set {#1,#P} will precede {#2,#3}, and so on. <i>Note. The term <u>initial ranking</u> always refers to the definition in section C.04.2.B, stating that the highest ranked player is first and the lowest ranked player is last.</i>
D.1	Sorting the upfloaters <i>All those players that have a lower score than the resident players of the scoregroup to be paired, are possible upfloaters and constitute the selected pool (see D.0.a).</i> 1. Main criterion Each possible upfloater receives a sequence number, according to their score and, when scores are equal, to their initial ranking. 2. Sets of upfloaters Because a set of upfloaters may be formed of players with different scores, all the possible sets are subdivided in containers. Sets belong to the same container if their players have the same scores.

		<i>Example:</i> <i>Let's assume that #1,#2,#3 have 3 points, #4 and #5 have 2.5 points, and #6 has 1.5 point, and a set of two upfloaters is needed. Then {#1,#2} {#1,#3} {#2,#3} are part of the same container; {#1,#4} {#1,#5} {#2,#4} {#2,#5} {#3,#4} {#3,#5} are part of another container; {#1,#6} {#2,#6} {#3,#6} are part of a third container; {#4,#5} are part of a fourth container; {#4,#6} {#5,#6} are part of a fifth (and last) container.</i>	
	3.	The containers are sorted along the lines described by criterion C.6. The sets belonging to each container are sorted according to the lexicographic order of the sequence numbers they are formed of.	
	D.2	Sorting the shifters <i>Any player in the bracket having a colour preference for White (Black) is a possible White (resp. Black) shifter. The need for shifters arises when, in order to make or complete a pairing, some players seeking a colour are shifted to the subgroup of players initially seeking the other colour.</i> <i>The possible White (resp. Black) shifters constitute the selected pool (see D.0.a).</i> 1. White seekers are sorted in order of ascending ARO or, when AROs are equal, highest initial ranking. Black seekers are sorted according to their initial ranking. 2. With such sorted list, assign the sequence numbers, starting with the player in the (remaining) middle of the list or, when two players are in the (remaining) middle, to the one with a higher position in the list. <i>Example:</i> <i>if the sorted list contains seven players (in order: A, B, C, D, E, F, G), #1 goes to D (middle of the seven players), #2 to C (higher between C and E, both in the middle of the remaining six players), #3 to E (middle of the remaining five players), #4 to B, #5 to F, #6 to A, #7 to G.</i> 3. Rationale: Since the system tries to equalize the ARO of the White seekers (while the Black seekers are “tools” for reaching this goal), it is statistically better to shift White seekers with AROs in the middle (<i>their ARO is probably already equalized</i>), and Black seekers with ratings in the middle (<i>because ARO equalization is usually performed better by Black seekers with extreme ratings</i>).	
	4.		
	D.3	Sorting G2 players (transpositions) <i>The players involved are the ones that end up in the G2 subgroup after the maneuvers</i>	

		<i>described in article B.4.</i> <i>Such players constitute the selected pool (see D.0.a).</i> 1. The players in the G2 pool are assigned sequence numbers according to their initial ranking. The sorted sets of G2 players are also called Transpositions. <i>Note:</i> <i>If, for instance, players A, B, C (listed according to the initial ranking) are in G2, the different Transpositions are {A, B, C} {A, C, B} {B, A, C} {B, C, A} {C, A, B} and {C, B, A}, in that exact order.</i>	
	2.		
E	Colour Allocation rules <i>Initial-colour</i> It is the colour determined by drawing of lots before the pairing of the first round. For each pair apply (with descending priority):		
	E.0	When both players have yet to play a game, if the higher ranked player (<i>the player who has more points or, when points are equal, a higher position in the initial ranking list</i>) has an odd pairing number, give him the initial-colour; otherwise give him the opposite colour. <i>Note:</i> <i>Always consider sections C.04.2.B/C (Initial Order/Late Entries) for the proper management of the pairing numbers.</i>	
	E.1	Grant both colour preferences.	
	E.2	Grant the stronger colour preference.	
	E.3	Taking into account C.04.2.D.5, alternate the colours to the most recent time in which one player had white and the other black.	
	E.4	Grant the colour preference of the higher ranked player (<i>see E.0</i>).	

C.04.4.2 Burstein System

Preface:

The BURSTEIN Swiss Pairing System is designed to maximize the fair treatment of players – in the sense that players having the same score should have met an average opposition as equal as possible during a tournament.

The system evaluates the strength of the opposition by means of an Index that only uses current data of the tournament, and is based on tie-break derived method(s) (Ratings are taken into account only when everything else is equal – see articles A.6-A.7). If this Index gives a nearly equal evaluation of all players in the same scoregroup, the goal is reached. Nevertheless, since a Swiss System is a more or less statistical system, this goal can only be reached approximately.

The approach is to attempt to equalize the Index of all players in a given scoregroup. Once the system is properly seeded (by pairing a number of early rounds using traditional methods – see A.5), the Index becomes a good evaluator of players' strength. Henceforth, in each round, the system will try to pair players who have a high Index with players who have a low Index within the same scoregroup. Although the immediate effect of this manoeuver is negligible (the Index is based on opponents' scores, and the players paired together often have equal scores), in the long run the desired effect is achieved (approximately, as mentioned above).

A.	Introductory Remarks and Definitions		
		A. 1	Initial ranking list
			See C.04.2.B (General Handling Rules – Initial order)
		A. 2	Scoregroups and pairing brackets
			A scoregroup is composed of all the players with the same score. A (pairing) bracket is a group of players to be paired. It is composed of players coming from a scoregroup (called resident players) and (possibly) of players who remained unpaired after the pairing of the previous bracket (called incoming floaters).
		A. 3	Byes
			See C.04.1.c (<i>Should the number of players to be paired be odd, one player is unpaired. This player receives a pairing-allocated bye: no opponent, no colour and as many points as are rewarded for a win, unless the regulations of the tournament state otherwise</i>).
		A. 4	Colour differences and colour preferences
		The colour difference of a player is the number of games played with white minus the number of games played with black by this player. The colour preference (<i>also called: due colour</i>) is the colour that a player should ideally receive for the next game. It can be determined for each player who has played at least one game.	
		a.	An absolute colour preference occurs when a player's colour difference is greater than +1 or less than -1, or when a player had the same colour in the two last rounds he played. The preference is white when the colour difference is less than 1 or when the last two games were played with black. The preference is black when the colour difference is greater than +1, or when the last two games were played with white.
		b.	A strong colour preference occurs when a player's colour difference is +1 (preference for black) or 1 (preference for white).
		c.	A mild colour preference occurs when a player's colour difference is zero, the preference being to alternate the colour with respect to the previous game he played.
		d.	Players who did not play any games have no colour preference (the preference of their opponents is granted).
		A.5	Seeding Rounds
		In order to properly seed the system, some initial rounds, called seeding rounds, are paired following the rules of the FIDE (Dutch) System (<i>see C.04.3</i>).	

		The number of seeding rounds is equal to half the number of rounds in the tournament (rounded down) or 4 (four), whichever is lower.	
		A.6	Opposition Evaluation
		During the pairing process, the players in a bracket need to be sorted (<i>see B.0.e, E.0, E.4, or section D</i>), by applying some or all of the methods defined here, as directed by article A.7.a.	
		a.	Sorting Methods
			1 Buchholz
			.
			It is the sum of the (current) scores of the opponents the player met.
			2 Sonneborn-Berger
			.
			It is the sum of the products given by the points the player earned against each opponent times the (current) scores of that opponent. <i>Note: If the standard scoring system is used, the above means the sum of the score of the opponents a player has defeated plus half the sum of the score of the opponents with whom he has drawn.</i>
		b.	Common Rules
			1 Unplayed games
			.
			If a player does not play in a round, the round shall be considered as one in which that player played against himself getting the result (win, draw, loss) that yields the same number of points as registered for the standings (and the future pairings). Exception: if a player has a series of consecutive zero-point-byes up to the current round, each of the ones gathered in previous rounds, for the benefit of the player's actual over-the-board opponents, is considered as a draw.
			2 Acceleration methods
			.
			If virtual points are used (<i>e.g. with the Baku Acceleration Method – see C.04.5.1</i>), such virtual points shall be excluded from the computation of any method.
		A.7	Ranking Order
		After the seeding rounds (<i>see A.5</i>), for pairings purposes only, the players in a bracket are ranked in order of, respectively:	
		a	(Opposition Evaluation) Index, which is a sequence of the methods seen in A.6.a, to be applied in the following order (<i>any subsequent method is used when preceding method(s) yield equal values</i>):
			1. Buchholz (<i>see A.6.a.1</i>)
			2. Sonneborn-Berger (<i>see A.6.a.2</i>)
		b	pairing numbers assigned to the players according to the initial ranking list (<i>see A.1</i>) and subsequent modifications depending on possible late entries or rating adjustments.

		Note: Players' scores are not used in the pairing ranking order.	
		A.8	RoundPairing Outlook
		Note This outlook is valid after the seeding rounds (see A.5) have been completed. The pairing of a round (called round-pairing) is complete if all the players (except at most one, who receives the pairing-allocated bye) have been paired and the absolute criteria C1C3 have been complied with. The pairing process starts with the assignment of the pairing-allocated-bye (<i>see B.0</i>) and continues by pairing the top scoregroup, and then bracket by bracket until all the scoregroups, in descending order, have been used and the round-pairing is complete. If it is impossible to complete a round-pairing, the arbiter shall decide what to do.	
		Section B describes the pairing procedures. Section C defines all the criteria that the pairing of a bracket has to satisfy (in order of priority). Section E defines the colour allocation rules that determine which players will play with White.	
B.	Pairing Procedures		
	Pairing-Allocated-Bye assignment		
	B.0	The pairing-allocated-bye is assigned to the player who:	
		a.	has neither received a pairing-allocated-bye, nor scored a forfeit win in the previous rounds (see C.2)
		b.	allows a complete pairing of all the remaining players (see C.4)
		c.	has the lowest score
		d.	has played the highest number of games
		e.	occupies the lowest ranking (<i>according to A.7</i>)
	Pairing Process for a bracket		
	B.1	The pairing of a bracket is composed of pairs and outgoing floaters. Determine the maximum number of pairs that can be obtained in the current bracket while complying with criteria from C.1 to C.5. This automatically determines the number of outgoing floaters.	
	B.2	Choose the first pairing (as ordered according to D.1) that complies best with all the pairing criteria (C.1 to C.8). Consider that a pairing is better than another if it better satisfies a quality criterion (C.5-C.8) of higher priority.	
C.	Pairing Criteria		
	Absolute Criteria		
	No pairing shall violate the following absolute criteria:		
	C.1	see C.04.1.b (<i>Two players shall not play against each other more than once</i>)	

	C.2	see C.04.1.d (<i>A player who has already received a pairing-allocated bye, or has already scored a (forfeit) win due to an opponent not appearing in time, shall not receive the pairing-allocated bye</i>).
	C.3	two players with the same absolute colour preference (<i>see A.4.a</i>) shall not meet (<i>see C.04.1.f and C.04.1.g</i>).
Completion Criterion		
	C.4	choose the set of outgoing floaters in order to complete the round-pairing.
Quality Criteria		
	To obtain the best possible pairing for a bracket, comply as much as possible with the following criteria, given in descending priority:	
	C.5	maximize the number of pairs (<i>equivalent to: minimize the number of outgoing floaters</i>)
	C.6	first maximize the number and then the scores of the incoming floaters that can be paired.
	C.7	choose the outgoing floaters so that in the following bracket C.4, C.5 and C.6 are complied with in the best possible way (<i>i.e., besides C.4 compliance, the number of pairs must be maximized, and all these floaters must be paired – or, at least, the ones with the highest scores</i>).
	C.8	minimize the number of players who do not get their colour preference.
D.	Order of pairings	
	All players in the bracket shall be tagged with consecutive in-bracket sequence-numbers (BSN for short) representing their respective ranking order (<i>according to A.7</i>) in the bracket (<i>i.e. 1, 2, 3, 4, ...</i>). The bracket is then extended, adding a number of virtual players equal to the number of outgoing floaters (<i>see C.5</i>). All those virtual players are assigned a BSN equal to zero, meaning that their opponent shall float.	
	D.1	In order to sort all the possible pairings, apply the following rule: a pairing precedes another if its BSN #1's opponent has a higher BSN (<i>i.e. lower ranking</i>) than the other's. If BSN #1's opponents are the same, then compare BSN #2's opponents; and so on.
E.	Colour Allocation rules	
	<i>Initialcolour</i> It is the colour determined by drawing of lots before the pairing of the first round.	
	For each pair apply (with descending priority):	
	E.0	When both players have yet to play a game, if the higher ranked player (<i>according to A.7</i>) has an odd pairing number, give him the initialcolour; otherwise give him the opposite colour. Note: Always consider sections C.04.2.B/C (Initial Order/Late Entries) for the proper management of the pairing numbers.
	E.1	Grant both colour preferences.

	E.2	Grant the stronger colour preference.
	E.3	Taking into account C.04.2.D.5, alternate the colours to the most recent time in which one player had white and the other black.
	E.4	Grant the colour preference of the higher ranked player (<i>see A.7</i>).

C.04.4.3 Lim System

<i>Approved by the General Assembly of 1987.</i>		
<i>Amended by the 1988, 1989, 1997, 1998 General Assemblies and 1999 Executive Board.</i>		
General Pairing Rules		
1	Awarding the pairing-allocated bye	
	1.1	In addition of what is stated in the basic rules (C.04.1.c), the pairing-allocated bye is awarded to the player with the lowest rank in the lowest score-group.
2	Pairing a Score-group	
	2.1	Two players who have not yet played each other are said to be compatible provided that the pairing will not require either player to have the same colour in three successive rounds, or to have three more of one colour than the other.
	2.2	The players with the same score form a score-group. The Median Score-group is the score-group with players having the score equal to half the number of rounds that have been played. Pairing begins with the highest score-group and proceeds downward until just before the Median Score-group, then continues with the lowest score-group and proceeds upwards to the Median Score-Group which is paired last. The Median-Score-group is paired downward.
	2.3	Before the players in a score-group are paired, the players in the score-group who have no suitable opponents for the following reasons are identified and transferred to a neighbouring score-group:
	a.	the player has already played all the players of his score-group; or
	b.	the player has already received two more of one colour over an equal allocation and there is no compatible opponent available in the score-group to enable him to have a permissible colour; or
	c.	the player has already received the same colour in the previous two rounds and there is no compatible player in the score-group to enable the player to have the alternate colour; or
	d.	it is necessary to make even the number of players in the score-group
		Such a transferred player is described as a floater. Rules on how to select the floater, if a choice is available, are given in the section on "Floater Selection Rules" (section 3).
	2.4	The players in a score-group, after transfer of players where necessary, are arranged in the order of their pairing numbers and the players in the top half are tentatively paired with the players in the bottom half. These pairings are said to be proposed pairings, to be confirmed after scrutiny for compatibility

		and proper colour. If the players in a score-group are numbered : 1, 2, 3 ... n, then the proposed pairings are (ignoring colours): 1 v (n/2 + 1), 2 v (n/2 + 2), 3 v (n/2 + 3) ... n/2 v n.
	2.5	Where a proposed pairing would result in the pairing of players who have already played each other, the lower numbered player of the two is exchanged for another within the same score-group. Further exchanges of opponents may be made to allow alternation or equalisation of colours where possible. How players are exchanged is described in the "Exchange Rules" (section 4).
	2.6	Pairing a blocked median score-group If the median score-group cannot be paired it should be extended step by step under the following rules: • if the number of floaters from higher score-groups is larger than the number of floaters from lower score-groups the next pairing of the lower score-group shall be cracked and the players of this pairing shall be treated as additional floaters from the lower score-group. Then the pairing of the median score-group is started again. • if the above condition is not fulfilled, then the next pairing of the higher score-group shall be cracked and the players of this pairing shall be treated as additional floaters from the higher score-group. Then the pairing of the median score-group is started again.
3	Floater Selection Rules	
	3.1	The "floater" is a player who is transferred to another score-group in accordance with Rule 2.3, or because a compatible opponent cannot be found for the player in spite of exchanges in the score-group.
	3.2	When pairing proceeds downward, the floater is transferred to the next lower score-group. When pairing proceeds upwards, the floater is transferred to the next higher score-group. When making even a score-group, determine the due colours of the players and select as the floater a player who would tend to equalise the number of players due different colours. (In Maxi-tournaments, when pairing downward, the difference in rating between the chosen player and the lowest numbered player in the score-group must differ by 100 points or less, otherwise the lowest numbered player in the score-group is chosen as the floater. When pairing upwards, the difference in rating between the player chosen and the highest numbered player in the score-group must differ by 100 points or less, otherwise the highest numbered player is chosen as the floater.) If the number of players due white equals the number of players due black, the lowest numbered player is chosen as the floater when pairing downward, and the highest numbered player is chosen as the floater when pairing upwards.
	3.3	If there is a choice as to which player floats to a lower score-group, the player chosen is the lowest numbered player in the score-group who has a compatible opponent in the lower score-group, after excluding the

		opponents of other floaters who have higher scores or higher pairing numbers than the proposed floater.
	3.4	If there is a choice as to which player floats to a higher score-group, the player chosen is the highest numbered player in the score-group who has a compatible opponent in the higher score-group, after excluding the opponents of other floaters who have lower scores or lower pairing numbers than the proposed floater.
	3.5	If a proposed floater has no compatible opponent in the adjacent score-group, he shall, if possible, be exchanged for another player in his score-group; otherwise he shall be floated to a further score-group.
	3.6	When pairing a group that includes floaters from a higher score-group, the floater with the highest score is paired first, or the floater with the highest pairing number, if scores are equal.
	3.6.1	When pairing a group that includes down-floaters (DF) from a higher score-group, the floater with the higher pairing number is paired first.
	3.6.2	When pairing a group with DF coming from different higher score-groups, the floater coming from the highest score group is paired first (not always the one with the highest pairing number).
	3.6.3	When there are DF and UF (up-floaters) in the same score-groups (this should normally happen in the median score-group) in the upper half of score-groups or in the median group, first pair the DF, then the UF and finally the remaining players.
	3.7	When pairing a group that includes floaters from a lower score-group, the floater with the lowest score is paired first, or the floater with the lowest pairing number, if scores are equal.
	3.7.1	When pairing a group that includes UF from a lower score-group (in the 2nd half) the floater with the lowest pairing number is paired first.
	3.7.2	When pairing a group that includes UF coming from different lower groups, the UF coming from the lowest score-group is paired first (not always the player with the highest pairing number).
	3.7.3	When there are UF and DF in the same score group in the second half of score-groups, first pair the UF, then the DF, and finally the other remaining players.
	3.8	When pairing downward, the floater is paired with the highest numbered player available who is due the alternate colour (provided, in Maxi-tournaments, that the ratings of proposed opponents who are exchanged for this purpose differ by 100 points or less). When pairing upwards, the floater is paired with the lowest numbered player available who is due the alternate colour (provided, in Maxi-tournaments, that the ratings of proposed opponents who are exchanged for this purpose differ by 100 points or less).
	3.9	Due to their origin and their compatibility in the adjacent score-group there are 4 types of floaters listed in descending order of disadvantages.

		a.	a floater who has already floated to the score-group just being handled and has no compatible opponent in the adjacent score-group.
		b.	a floater who has already floated to the score-group just being handled and has a compatible opponent in the adjacent score-group.
		c.	a floater who has no compatible opponent in the adjacent score-group.
		d.	a floater who has a compatible opponent in the adjacent score-group.
		If there is a choice the floaters should be chosen to minimise the disadvantages using the following priorities:	
		a.	avoid floater(s) of type a
		b.	avoid floater(s) of type b
		c.	avoid floater(s) of type c
	3.10	A floater who has floated the round just before shall not be floated due to section 2.3.d provided: • this will not produce other floaters of the types a, b, c of section 3.9 • this will not decrease the number of pairings of that score-group	
4.	Exchange Rules		
	4.1	The proposed pairings of players obtained according to Rule 2.4 are scrutinised in turn for compliance with the compatibility statement (see 2.1). And,	
		a.	when pairing downward, scrutiny of proposed pairings begins with the highest numbered player; if the pairing is found not to comply with 2.1, the lower numbered player is exchanged until a compatible pairing is found; or,
		b.	when pairing upwards, scrutiny of proposed pairings begins with the lowest numbered player; if the pairing is found not to comply with 2.1, the higher numbered player is exchanged until a compatible pairing is found.
	4.2	In the following example of a score-group with six players, and pairing downward, the attempt is first made to find a compatible opponent for Player #1, the highest numbered player in the score-group. Six players in a score-group with proposed pairings as follows: 1 v 4 2 v 5 3 v 6 If the pairing 1 v 4 is not compatible, for example, because the players had met in an earlier round, the positions of Player #4 and Player #5 are exchanged so that we have: 1 v 5 2 v 4 3 v 6 If the pairing 1 v 5 is also not compatible, a further exchange is made. The	

		original proposed pairing and possible exchanges made to find a compatible opponent for Player #1 are as follows: Proposed Pairing (col. 1) and Possible exchanges to find compatible opponent for #1																			
		<table><tr><td>1 v 4</td><td>1 v 5</td><td>1 v 6</td><td>1 v 3</td><td>1 v 2</td></tr><tr><td>2 v 5</td><td>2 v 4</td><td>2 v 4</td><td>2 v 5</td><td>3 v 5</td></tr><tr><td>3 v 6</td><td>3 v 6</td><td>3 v 5</td><td>4 v 6</td><td>4 v 6</td></tr></table>					1 v 4	1 v 5	1 v 6	1 v 3	1 v 2	2 v 5	2 v 4	2 v 4	2 v 5	3 v 5	3 v 6	3 v 6	3 v 5	4 v 6	4 v 6
1 v 4	1 v 5	1 v 6	1 v 3	1 v 2																	
2 v 5	2 v 4	2 v 4	2 v 5	3 v 5																	
3 v 6	3 v 6	3 v 5	4 v 6	4 v 6																	
	4.3	After a compatible opponent, for example, #6, has been found for Player #1, the proposed pairing for Player #2 is scrutinised. Exchanges to find a compatible opponent for Player #2 are as follows: Proposed Pairing (col. 1) and Possible exchanges to find compatible opponent for #2																			
		<table><tr><td>1 v 6</td><td>1 v 6</td><td>1 v 6</td><td>1 v 3</td><td>1 v 2</td></tr><tr><td>2 v 4</td><td>2 v 5</td><td>2 v 3</td><td>2 v 6</td><td>3 v 5</td></tr><tr><td>3 v 5</td><td>3 v 4</td><td>4 v 5</td><td>4 v 5</td><td>4 v 6</td></tr></table>					1 v 6	1 v 6	1 v 6	1 v 3	1 v 2	2 v 4	2 v 5	2 v 3	2 v 6	3 v 5	3 v 5	3 v 4	4 v 5	4 v 5	4 v 6
1 v 6	1 v 6	1 v 6	1 v 3	1 v 2																	
2 v 4	2 v 5	2 v 3	2 v 6	3 v 5																	
3 v 5	3 v 4	4 v 5	4 v 5	4 v 6																	
	4.4	The exchanges to find a compatible opponent for Player #2 must at the same time leave Player #1 with a compatible opponent. If this cannot be done, for example, if Player #1 and Player #2 have previously played each other and all the other players except Player #6, then the original pairing of Player #1 with Player #6 is retained and Player #2 is floated. And,																			
		a.	if the score-group originally had uneven members and the lowest numbered player was floated to make even the number of players in the score-group, #2 is exchanged with the floater, originally #7 in the score-group, or,																		
		b.	if the score-group was originally even, then the lowest numbered player remaining must be floated in company with #2 to maintain an even number of members in the score-group.																		
5. Colour allocation rules																					
	5.1	Where possible, and by means of exchanges, each player shall be given the alternate colour; at the end of each even-numbered round each player shall have had an equal number of whites and blacks. Moreover,																			
		a.	no player shall be given the same colour in three successive rounds, and																		
		b.	no player shall be given three more of one colour than the other.																		
	5.2	After the first scrutiny and exchanges necessary to establish that all pairings in a score-group are new pairings, a second scrutiny with exchanges where necessary is undertaken to give each player, if possible, the alternating colour and at the same time, the equalising colour.																			
	5.3	If one of the players in a pairing had the same colour in the previous two rounds, he must be given the alternating colour. If both players had the same colour in the previous two rounds and compatible opponents in the score-group are not available, then one or both players must be floated.																			

	5.4	If both players in a pairing had the same colour in the previous round, then the colours they had in earlier rounds, going back in sequence, shall decide who is given the alternate colour. If players in the median score-group or above had identical histories, then the higher ranked is given the alternate colour, or, in even-numbered rounds, the equalising colour. If the players below the median score-group had identical histories, then the lower ranked player is given the alternate colour, or, in even numbered rounds, the equalising colour.
	5.5	In the odd-numbered rounds, whenever possible, each player shall be given the colour which gives him one more only of one colour than the other.
	5.6	In the even-numbered rounds, whenever possible, each player shall be given the colour that gives him an equal number of whites and blacks. When both players of a pairing are due the same equalising colour, and further exchanges are not possible, the colour history will decide who is given the equalising colour, as in Rule 5.4. One player will then have two more of one colour than the other colour. This is allowed but care must be taken not to violate Rules 5.1(a) and 5.1(b), and to equalise the player's colours at the earliest opportunity.
	5.7	(In Maxi-tournaments, an exchange of opponents to find, for example, one who is due the alternate colour is allowed only if the ratings of the opponents to be exchanged differ by 100 points or less.)
6.	Exceptions applicable to the last round. In the last round, the general principle C.04.1.e, requiring players with the same score to be paired if they had not met in an earlier round, shall have priority over alternation and equalisation of colours, even if it is necessary for one of the players to be given the same colour for the third round in succession, or to be given three more of one colour than the other.	
Brief examples of pairing		
7.	Pairing Round One	
	7.1	If the number of players is uneven the lowest rated player in the Pairing List is given the pairing-allocated bye.
	7.2	The colour to be given to Player #1 is decided by drawing lots; the other odd-numbered players in the upper half of the Pairing List are then given the same colour as Player #1. Player #2 together with the other even-numbered players in the upper half of the Pairing List are given the other colour. Depending on the draw, the pairings for the first round in a tournament of forty players would be either 1 v 21, 22 v 2, 3 v 23, 24 v 4, ... 40 v 20; or 21 v 1, 2 v 22, 23 v 3, 4 v 24 ... 20 v 40, where the player having white is mentioned first. This is the only occasion when colours need be decided by lot.
	7.3	Players who have won their games are each awarded one point; each of those who have drawn receives 0.5 point. Each of those who have lost receives 0 point.
8.	Round Two	

	8.1	The players are arranged in groups of the same score.
	8.2	If the number of players is uneven, then the pairing-allocated bye is awarded as in Rule 1.
	8.3	Pairing begins with the highest score-group (1 point), continues with the lowest score-group (0 point) and finishes with the Median Score-group (0.5 point). Detailed instructions for pairing Round Two and subsequent rounds are above.

C.04.5 FIDE-approved Accelerated Systems

C.04.5.1 Baku Acceleration

1.	Premise	
	The Baku Acceleration Method is applicable in any tournament where the standard scoring point system (one point for a win, half point for a draw) is used.	
2.	Initial Groups Division	
	Before the first round, the list of players to be paired (properly sorted) shall be split in two groups, GA and GB. The first group (GA) shall contain the first half of the players, rounded up to the nearest even number. The second group (GB) shall contain all the remaining players.	
	<i>Note:</i>	<i>for instance, if there are 161 players in the tournament, the nearest even number that comprises the first half of the players (i.e. 80.5) is 82. The formula $2 * Q$ (2 times Q), where Q is the number of players divided by 4 and rounded upwards, may be helpful in computing such number – that, besides being the number of GA players, is also the pairing number of the last GA player.</i>
3.	Late entries	
	If there are entries after the first round, those players shall be accommodated in the pairing list according to C.04.2.B/C (Initial Order/Late Entries). The last GA-player shall be the same as in the previous round.	
	<i>Note 1:</i>	<i>In such circumstances, the pairing number of the last GA-player may be different by the one set accordingly to Rule 2.</i>
	<i>Note 2:</i>	<i>After the first round, GA may contain an odd number of players.</i>
4.	Virtual points	
	The “accelerated rounds” are the ones in the first half (rounded up) of the tournament. Before pairing the first half (rounded up) of the accelerated rounds, all the players in GA are assigned a number of points (called virtual points) equal to 1. Such virtual points are reduced to 0.5 before pairing the remaining accelerated rounds.	

	<i>Note:</i>	<i>Consequently, no virtual points are ever given to players in GB, or to any player after the last accelerated round has been played.</i>
	Example:	In a nine-round tournament, the accelerated rounds are five. The players in GA are assigned one virtual point in the first three rounds, and half virtual point in the next two rounds.
5.	Pairing score	
	The pairing score of a player (i.e. the value used to define the scoregroups and internally sort them) is given by the sum of his standings points and the virtual points assigned to him.	

C.04.A Appendix: Endorsement of a software program

A.1	Introduction
	To manage big Swiss tournaments, the use of computer programs, to handle players' data, pairings and results, is necessary. To avoid misunderstandings and to support both the organisers and the arbiters, FIDE recommends the use of computer programs endorsed by FIDE itself, after a successful endorsement procedure.
A.2	The endorsement procedure for a pairing system
	Each author of a program that helps to manage a chess tournament can apply for the FIDE endorsement by submitting an FE-1 form (<i>see Annex-1</i>). For an endorsement application to be considered, the program must be able to manage Swiss tournaments using the FIDE (Dutch) System (<i>see C.04.3</i>) or any other pairing systems approved by FIDE (<i>see C.04.4.1-3</i>). The endorsement is given for the specific pairing systems (one or more). Any program asking for endorsement should provide (explicitly or implicitly) a FIDE mode, which should offer all the functionalities and services required by FIDE for a tournament-managing program to be endorsable (<i>see below</i>). The program is to be endorsed in the FIDE mode. Moreover, it must provide the following services: • an English language interface • the capability to import and export files coded in the FIDE Data Exchange Format (<i>see A.3.1 and Annex-4</i>) • the public availability of a (free) pairings checker (FPC – see A.4) • the public availability of a (free) generator of simulated tournaments (RTG, see A.5), unless exempted by the System of Pairings and Programs Commission (SPPC) • the possibility to be checked in a controlled environment • the compliance with all the requirements presented in the Verification Check List (<i>see Annex-4</i>) <i>The applicant should consider that merely complying with all the aforementioned requirements is not enough to receive a FIDE endorsement.</i>

	The FIDE mode may also offer additional services or functionalities, provided that they are not explicitly prohibited by FIDE, on condition that those services and functionalities may not cause pairing mishaps for FIDE mode users. If, during the period of validity of the endorsement (<i>see A.8</i>), a breach of the above conditions is reported to the SPPC, and verified by the Commission, the endorsement may be immediately suspended (pending further investigation) or permanently revoked. In the latter case, the program reverts to the status of a new program to endorse. If an error is discovered or reported in an endorsed software program, the secretary of the “Systems of Pairings and Programs” will send a notification to the supplier of the program to correct the error. Errors will be classified as major or minor. Major errors must be fixed within two weeks after from the time the secretary send the notification and within two months for minor errors. Should the error not be fixed within the stipulated timeframe, the endorsement of the programs will be automatically suspended until the error is fixed to the satisfaction of the “Systems and Pairings Committee Council	
A.3	Data Exchange formats and files	
	A.3.1	FIDE Data Exchange Format
		See Annex-2/TRF16 or TRF06 for the descriptions of the current FIDE Data Exchange Format (<i>version 2016, also called TRF16</i>) or the old one (<i>version 2006, also called TRF06</i>).
	A.3.2	Tournament Report File (TRF)
		It is a file coded in TRF16 or TRF06.
A.4	(Free) Pairings Checker (FPC)	
	An External Pairings Checker is a tool, embedded in the main program and containing the pairing engine, that can be freely used by anyone (without the user interface). For instance, be <i>yourprogram.exe</i> the name of the program executable. It is (normally) launched by a command prompt or double-clicking on it. The regular services it offers are not relevant from the (<i>External Pairings</i>) checker viewpoint. What it is expected from a checker is the possibility to launch <i>yourprogram.exe</i> from a command prompt with some parameters, like in: <i>yourprogram.exe -check FIDE_Report_File.fid</i> The checker must be able to read FIDE_Report_File.fid when coded in TRF16 and should be able to read it when coded in TRF06 (<i>within the limitations of such a format</i>). Then, for each round, the checker must rebuild the tournament, pair the round using the embedded pairing engine, and output a report describing which pairings are or are not consistent with those produced by the pairing engine.	
A.5	Random Tournament Generator (RTG)	
	The RTG is a freely available tool that, preferably run from a command prompt, can easily generate many simulated tournaments producing a full TRF (TRF16) file for each of them. In all those tournaments, the pairing rules implemented by the pairing engine must	

	be strictly followed. It is recommended that the games results may somewhat respect the probabilities given by the FIDE rating table (<i>see pairings.fide.com -Documents section-, for suggestions on how to code such a probability table</i>).	
A.6	The first endorsement procedure for a pairing system	
	A subcommittee of four people must be named by the SPPC at the first Congress that follows the application for the endorsement of a program, as long as such naming activity is inserted into the SPPC agenda. The subcommittee shall report to the next Congress whether the program is suitable to be endorsed.	
A.7	Endorsement for pairings systems when other programs have already been endorsed for the same pairing system	
	As the verification of the pairing engine can be performed by computerized methods, and all the tools needed for such verification are freely available to the authors, the endorsement request can reach the SPP secretariat at any moment, as long as this is at least four months before the Congress to which it would be presented. As, by definition, an external RTG is available, it will be used to generate 5000 random tournaments. Such tournaments will be given in input to the candidate FPC and each discrepancy, as long as there are at most 10 of them, will be collected. Such discrepancies may depend on either: • an error in the input file (i.e. they are the responsibility of the endorsed program which provided the RTG) <i>[or]</i> • an error coming from the candidate <i>[or]</i> • an interpretation divergence caused by unclear rules	
	Errors of the first type must be redirected to the RTG provider. Problems of the third type are redirected to the SPPC, which should issue a statement describing the proper interpretation, and then fix the problems in the successive revision of the rules. Errors of the second type must be corrected in a reasonable time-frame before the candidate is presented to the Congress for endorsement. If the candidate has its own RTG, the latter is used to generate 5000 random tournaments, which will be then given in input to one (or more) of the available FPC(s). The analysis of the discrepancies is conducted in the same way as above.	
A.8	SPPC Evolution Plan	
	A.8.1	Endorsement Cycle
		It is a four-year period running from January 1st of YearX+1 to December 31st of YearX+4, where X is a leap year (for instance 2016, 2020, ...). Any endorsement given during the first three years of the Endorsement Cycle will last until the end of the Endorsement Cycle. No endorsement is given during the last year of the Endorsement Cycle, unless the SPPC decides otherwise. FE-1 applications can only be presented for the following Endorsement Cycle.
	A.8.2	Transition Period
		It is a time-period lasting from January 1st of YearX+1 (i.e. the year of the beginning of the Endorsement Cycle) to the Congress of the same year. It is the period during which the SPPC shall run the endorsement procedures of the programs that either were already endorsed or are presented for

		endorsement before the end of the previous Endorsement Cycle (i.e. December 31st of YearX).	
	A.8.3	Interim Certificate	
		Any program that passes the endorsement procedure during the Transition Period receives an <i>interim certificate</i>, which will allow the program to be immediately used with FIDE approval. A list of the interim certified programs will be available on <i>pairings.fide.com</i> for the whole Transition Period. The ensuing Congress will then transform such certificate in an official endorsement, unless a formal complaint against the software is brought to the attention of the SPPC before the Congress itself.	
	A.8.4	List of FIDE endorsed programs (see also Annex 3)	
		The official endorsement of a program is decided at the Congress that follows the presentation of the proper application. Any program that is officially endorsed is inserted into a list of endorsed programs. Such list is an annex to this section and, consequently, is part of the FIDE Handbook. An endorsed program shall be removed from the above list if it fails to pass the endorsement procedure run during the Transition Period, and the ensuing Congress (i.e. the Congress of YearX+1) acknowledges the failure.	
	A.8.5	Rule Amendments	
		If there is any variation or major clarification of the pairing rules, the SPPC shall define such changes at last for the Congress of YearX+3. This kind of changes normally comes into force on July 1st of YearX+5. However (see rule A.8.3) interim certified programs may already use the amended rules as soon as they are endorsed during the Transition Period of YearX+5. During the Transition Period, if there is a theoretical possibility for a conflict, the tournament directors shall communicate to the players which version of the rules they are going to use in the tournament.	
A.9	Currently endorsed programs		
	See Annex-3, containing the list of the endorsed programs and useful information related to them.		
A.10	Section annexes		
	Annex-1	Application for Swiss Pairing Program FIDE Endorsement	FE-1
	Annex-2	Tournament Report File Format (version 2006)	TRF06
		Tournament Report File Format (version 2016)	TRF16
	Annex-3	List of FIDE Endorsed Programs	FEP19
	Annex-4	Verification check-list	VCL19